Tax Smoothing with Financial Instruments

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The paper analyzes the optimal structure of government debt in a stochastic environment. In a model with distortionary taxes, the government should smooth tax rates over states of nature as well as over time. Government liabilities should be structured to hedge against macroeconomic shocks that affect the government budget. The optimal structure of government liabilities generally includes some "risky" securities which are state-contingent in real terms. The empirical part of the paper tests for tax smoothing and then studies state contingencies implemented by some specific securities including nominal debt, long-term bonds, equity, and foreign-currency debt. (*JEL* 321)

The United States government issues Treasury bonds and bills of various maturities. They are considered risk-free in terms of default risk, though their real value may fluctuate considerably. This paper is concerned with questions of what may motivate such a debt structure and whether it is an optimal one.

The paper analyzes government policies that maximize welfare. The welfare-maximizing approach of analyzing government debt policy was introduced by Robert Barro (1979). He shows that, in a deterministic environment, optimal tax and debt policy should smooth tax rates over time. Optimal policy in a stochastic environment calls for state-contingent tax rates that must be supported by state-contingent debt, as Robert Lucas and Nancy Stokey (1983) have shown. I consider a stochastic version of Barro's model. Key assumptions (discussed in more detail below) are that welfare losses due to distortionary taxation can be summarized by a convex function of tax rates and that asset prices and return distributions are exogenous.

Optimal policy will then smooth tax rates over time and over states of nature. If there are macroeconomic shocks that affect the government budget, government liabilities should provide a hedge against these shocks. This characterizes the optimal structure of government liabilities.

If markets are complete and if the government can trade on all markets, the model has the very strong implication that changes in tax rates should never occur, because government liabilities could be contingent on all conceivable shocks. However, markets may be incomplete for various informational or incentive reasons, and even if they were complete, the government may not be able to operate on all markets.[1] Definitive answers on why certain markets are missing

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[1]See Franklin Allen and Douglas Gale (1988) on market incompleteness and Gale (1990) on the possibility that innovative debt management may open new markets. Here, the critical issue is not missing markets per se, but government access to markets. Government activity on markets designed to provide hedges against budgetary uncertainty may be particularly problematic, because of incentive and asymmetric-information problems. For example, the government might be able to reduce the volatility of taxes by issuing securities contingent on government spending or on the tax rate itself. However, if such securities were issued, the government would have an incentive to manipulate their payoffs by changing spending or taxes. One cannot claim, though, that incentive problems provide a full explanation for imperfect tax smoothing, since nominal debt is traded even though it clearly creates a time-consistency problem. Gale's point that innovative debt management may improve welfare is very much consistent with this paper (see Section IV-C), but to be cautious, I will only consider currently existing securities.

and on whether or not the government may trade on a certain market are beyond the scope of this paper. Nonetheless, unless one can rule out any incompleteness or incentive problems, it seems too much to require that the government stabilize tax rates perfectly. Therefore, I take the set of available securities (bundles of state-contingent claims) considered for the government portfolio as exogenous.

For any given set of securities, one can compute the optimal portfolio of government liabilities formed with these securities. The optimality conditions are that the tax rate is uncorrelated with the return on each security. For several well-known and widely traded securities (short- and long-term dollar bonds, stocks, foreign exchange, and foreign bonds), I test the zero-correlation condition and estimate optimal portfolios, making different assumptions about the set of available securities. The tests provide an assessment of tax smoothing as a positive theory of government. The estimated optimal portfolios in comparison with the actual structure of U.S. government debt show in which direction debt policy should be modified to improve tax smoothing. Since the estimation is more constructive and because of robustness considerations, the emphasis will be on the analysis of optimal portfolios, rather than on testing.

For the United States, several questions about the debt structure are interesting under positive as well as normative aspects. The first question relates to the absence of indexed debt (see, e.g., Stanley Fischer, 1983; Bohn, 1988). Nonexistence may be interpreted as indicating that a risk-free asset cannot be created, or it may be an equilibrium phenomenon, meaning that the government could issue indexed debt if it wanted to. The example of Britain suggests that indexed debt could be issued easily. The paper shows that, indeed, the specific state contingency implemented by relating the real value of government debt to inflation has desirable hedging properties. Thus, nonindexation is consistent with optimal policy.[2]

A second question is about the maturity structure of debt. In a discrete-time framework, the real value of debt with maturity greater than one period varies with nominal interest rates. I show that this type of contingency is also desirable for a welfare-maximizing government, though the evidence is weaker than that in favor of nominal debt.

Third, one may ask whether the government can improve on the current practice of issuing only domestic debt securities. Though an exhaustive survey of alternatives is beyond the scope of this paper, the two cases of stocks and foreign-currency debt are explored. One finds that the government could indeed hedge against shocks due to cyclical fluctuations by taking a short position in the stock market. Movements in some foreign interest rates also have desirable hedging properties, though there is little support for taking exchange-rate risk. Overall, it seems that the government could improve welfare by looking outside the set of dollar-dominated debt securities in structuring its liabilities.

The results on debt management rely heavily on the optimality of tax smoothing. Though the notion that excess burden increases on the margin with increasing tax rates is familiar from atemporal models of taxation, tax smoothing is not always optimal in dynamic models (see Olivier Blanchard and Fischer, 1989 Ch. 11.3; Richard Tresch, 1981 Part III). In general, excess burden may depend on other variables in addition to current tax rates (see, e.g., Lucas and Stokey, 1983). The tax-smoothing approach may work well for economies where taxes are labor income taxes or other taxes with largely static effects, but it is probably less appropriate for cases in which taxation has significant effects on interest rates or on capital accumulation (see Lucas and Stokey [1983] and Kenneth Judd [1989], respectively). In particular, if debt management affected interest rates, the qualitative nature of the government's optimization problem would change significantly, because the

[2]This abstracts from issues of time consistency, which would favor indexation (Guillermo Calvo, 1978; Lucas and Stokey, 1983; Bohn, 1988). Time-consistency issues have intentionally been omitted from the paper, because they would distract from the analysis of the government debt portfolio as a whole.

government would no longer behave as a price taker on financial markets. Thus, the normative results of this paper should apply to economies where debt management has few macroeconomic effects and where the current tax rate is the main determinant of excess burden, but they may have to be interpreted more cautiously otherwise.[3]

A final simplifying assumption is that individuals are risk-neutral. All expected returns are then tied to the rate of time preference, making them exogenous in a straightforward way. Somewhat surprisingly for a paper concerned with hedging, this assumption does not seem to affect the results; but it helps to focus on the government's problem. Regardless of the degree of individual risk aversion, the convexity of excess burden makes the government act "more risk-averse" than taxpayers and implies that tax smoothing is a close approximation to the optimal policy. Since the risk-neutral case shows most clearly how the government's hedging motive derives from the (added) concavity in the welfare function created by distortionary taxes, risk neutrality is imposed throughout (see the end of Section I for more details).

The paper is organized as follows. Section I sets up a simple model that provides a tax-smoothing argument for why governments may want to hedge against economic uncertainty. Tests for tax smoothing are in Section II. Section III derives an equation for the optimal liability structure, and Section IV contains estimates of optimal portfolios containing nominal, long-term, and foreign-currency bonds and stocks. Section V summarizes the results.

I. A Framework for Analysis

Barro (1979) has shown that, with distortionary taxes, the government should smooth tax rates over time. This section shows that Barro's approach generalizes in a stochastic environment to tax smoothing over states of nature. The government behaves as if it is averse to the risk of changing tax rates, even if all individuals are risk-neutral.

I consider a model similar to Barro (1979), except that risky securities are added. In period t, identical, infinitely lived individuals maximize

$$U_t = E_t \sum_{j \geq 0} \rho^j c_{t+j} \tag{1}$$

where $0 < \rho < 1$ is a discount factor and c_{t+j} is consumption in period $t+j$. They own a stream of endowments Y_{t+j} and may trade $K+1$ assets. Let $A_{k,t}$ be the quantity of asset k $(k = 0, 1, \ldots, K)$ purchased in period t, $p_{t,k}$ be the price of asset k in terms of consumption goods (ex dividend), and $f_{t+j,k}$, $j \geq 1$, be the stream of cash flows (interest payments or dividends) in future periods. Let returns be denoted by $r_{t+1,k} = (p_{t+1,k} + f_{t+1,k})/p_{t,k} - 1$.

Individuals pay taxes on endowments at a rate τ_t. I assume that taxes are distortionary (e.g., because of wasteful efforts of evading or sheltering income). Following Barro (1979), the excess burden of taxation is summarized by a loss function $h(\tau_t)$, which indicates the fraction of endowment "wasted" when taxes are τ_t. Then, the individual budget constraint is

$$c_t + \sum_k p_{t,k} A_{t,k} = Y_t[1 - \tau_t - h(\tau_t)] + \sum_k (p_{t,k} + f_{t,k}) A_{t-1,k}. \tag{2}$$

Individual optimization implies asset-pricing equations $p_{t,k} = \rho E_t(p_{t+1,k} + f_{t+1,k})$ for all k. That is, all expected returns must be equal:

$$E_t(1 + r_{t+1,k}) = \frac{1}{\rho} \quad \text{for all } k. \tag{3}$$

It is convenient to introduce several specific securities that will be analyzed later. First, let $k = 0$ be a risk-free (in real terms, i.e., price-level-indexed) one-period security that has a price $p_{t,0} = 1$ and return $r \equiv 1/\rho - 1 > 0$ for all t. Then one can define excess returns $\hat{r}_{t+1,k} \equiv r_{t+1,k} - r$ on assets $k \geq 1$. Individual optimization can be summarized by $E_t \hat{r}_{t+1,k} = 0$.

[3]At least for the choice of maturities, the assumption that debt management has little macroeconomic effect seems empirically defensible; see Franco Modigliani and Richard Sutch (1966).

Second, I want to discuss assets with returns defined in terms of a nominal unit of account, money. However, I do not want to focus on asset-pricing issues specific to monetary models, nor do I want to discuss optimal monetary policy. While both issues are important topics in themselves, all I need here is a well-defined price level. Therefore, I will just assume that the price level P_t and the rate of inflation, $\pi_t = \log(P_t/(P_{t-1})$, follow some stochastic processes. These assumptions could be motivated more rigorously as a limit of a cash-in-advance model with "small" monetary sector.

Third, some securities may be denominated in a foreign currency. Given risk-neutrality of domestic individuals, the closed-economy market-clearing conditions are not essential. That is, the existence of "other" individuals abroad does not change the model significantly. Whenever necessary, I will therefore assume that payoffs of some securities may depend on variables defined within a foreign economy.

The government uses tax revenues $T_t = \tau_t Y_t$ to finance government spending G_t and to service the government debt. I assume that the government can issue arbitrary quantities $D_{t,k}$ of the securities k at the market price. The government budget constraint is

$$(4)\quad T_t = \tau_t Y_t = G_t + \sum_k (p_{t,k} + f_{t,k}) D_{t-1,k} - \sum_k p_{t,k} D_{t,k}.$$

Individual welfare can be written as a function of government policy by substituting (4) and (2) into the individual objective function and dropping irrelevant terms:[4]

$$(5)\quad U_t = E_t \sum_{j \geq 0} \rho^j \{ Y_{t+j} [1 - h(\tau_{t+j})] \}.$$

The government chooses tax rates and debt structure to maximize (5) subject to (4). In effect, the government objective is to minimize the expected present value of excess burden. The first-order conditions for optimal policy are

$$(6a)\quad E_t[h'(\tau_{t+1})] = h'(\tau_t) \quad \text{for all } k = 0$$

$$(6b)\quad \rho E_t[h'(\tau_{t+1})(1 + r_{t+1,k})] = h'(\tau_t) \quad \text{for all } k > 0.$$

To simplify, assume that excess burden is quadratic; that is, $h(\tau_t) = (h/2)\tau_t^2$ for some $h > 0$. Then (6a) implies tax smoothing over time, $E_t \tau_{t+1} = \tau_t$, as in Barro (1979), or in terms of stationary variables,

$$(7a)\quad E_t(\Delta\tau_{t+1}) = 0.$$

This also determines the path of total debt. Using (3), equation (6b) can be rewritten as

$$(7b)\quad E_t[\Delta\tau_{t+1}(1 + r_{t+1,k})] = 0 \quad \text{for all } k > 0.$$

Notice that this equation—though not (7a)—would hold even if it were impossible to issue a risk-free asset. Combining (7a) and (7b), one obtains

$$(8)\quad E_t(\Delta\tau_{t1}\hat{r}_{t+1,k}) = E_t(\hat{\tau}_{t+1}\hat{r}_{t+1,k}) = \mathrm{Cov}_t(\tau_{t+1}, r_{t+1,k}) = 0$$

where $\hat{\tau}_{t+1} = \tau_{t+1} - E_t\tau_{t+1}$ is the innovation in tax rates. That is, the government should stabilize taxes across possible states of nature. This implies a zero conditional covariance between taxes and returns on all available securities. The optimality conditions (7) and (8) will be tested in Section II. They implicitly characterize the optimal debt structure, since taxes are a function of debt policy through the budget constraint. This link will be made explicit in Section III.

At this point, a remark on the assumption of risk-neutrality may be appropriate. If preferences (1) were replaced by a time-

[4] To be exact, $\Sigma_k(p_{t,k} + f_{t,k})(A_{t-1,k} - D_{t-1,k}) - E_t\Sigma_{j \geq 0}\rho^j G_{t+j}$ should be added to the right-hand side of (5); but since this is an additive, exogenous term, it does not affect decisions.

additive concave utility function (with money and foreigners excluded for the purpose of this argument), the ratio of $t+1$ and t-dated marginal utilities would enter (3), as in Lucas (1978). Through a similar modification of (5), the same ratio of marginal utilities would enter equations (6)–(8). If one takes such a consumption-based capital-asset-pricing model as the maintained hypothesis for normative analysis, a variation of Rajnish Mehra and Edward Prescott's (1985) argument (based on the fact that consumption growth has low variance) can be used to show that changes in the marginal rate of substitution are quantitatively unimportant (details available from the author). Independently from the Mehra-Prescott argument, the fact that tax rates and consumption have low correlation in U.S. data (less than 0.05 in absolute value for the period 1954–87, using nondurables) suggests that risk-neutrality is a justifiable simplification. Risk aversion does not affect the basic intuition that, because of the convex excess burden in its objective function, the government should smooth tax rates.[5]

II. Optimal Debt

In this section, the first-order conditions for taxes, (7) and (8), are tested for quarterly postwar United States data. The basic sample periods are 1954:2–1987:4 for domestic series and 1973:1–1987:4 for international series. The measure of tax rates is the ratio of federal tax revenue to GNP.[6]

The use of quarterly data seems necessary to obtain reasonably precise estimates of the covariances, but it may be somewhat problematic if tax policy is set less frequently. (The optimal debt portfolios in the next section will turn out to be more robust.) If a quarterly revision of tax policy in light of news about all relevant variables is not feasible in practice (e.g., for technical reasons [tax laws specify annual payments] or because of legislative delays), significant correlations in quarterly data should not be interpreted as rejections of optimality but, rather, as an indication that tax smoothing could be improved, if it were possible to adjust tax laws more frequently. Still, equations (7) and (8) yield the most natural and direct tests of the tax-smoothing model, even if negative conclusions may have to be interpreted cautiously.

The random-walk condition (7a) will only be discussed briefly, since it has been tested before (see Chaipat Sahasakul, 1986; Barro, 1981). Over the postwar sample, the change in tax rates has an insignificant mean of 0.023 percent ($t = 0.59$) and small autocorrelations. The series was regressed on its own lags, on various lagged asset returns, and on monetary data (those used in Section IV). The hypothesis that changes in tax rates are unpredictable was not rejected in any of the tests; the details are therefore omitted. These results confirm Barro (1981), but they are in contrast to the results of Sahasakul, who finds evidence against tax smoothing for a sample period that includes World War II. However, if one looks more carefully, the results do not contradict Sahasakul: simple tax-smoothing models apparently have some problems in explaining World War II data (which is not the subject

[5]The reference to Mehra and Prescott (1985) leads to a more fundamental problem, however, because their point was that consumption-based asset pricing cannot explain observed risk premiums. If the Mehra-Prescott puzzle means that actual risk premiums are too high for no good reason, one may speculate whether the optimal debt portfolios should be biased toward assets that promise low expected returns. However, if the "true" model that justifies the observed high equity premiums is similar to the Lucas (1978) model in that the same risk factor (perhaps something other than marginal utility of consumption) appears in both consumer and government first-order conditions, the basic intuition will still apply.

[6]The reason for using this measure instead of, say, marginal federal income tax rates is that it is unclear what "the" marginal tax rate is on aggregate. Taxes are levied not just on income but also on other activities; rates differ across individuals; and tax laws contain a multitude of exemptions and exceptions. On aggregate, however, all taxes must be paid out of the available economic resources, GNP. A higher revenue:GNP ratio implies higher tax rates somewhere, no matter how tax laws are structured in detail. Therefore, I consider changes of the revenue:GNP ratio as indicators of changes in tax rates and excess burden. See Barro (1981) for further discussion.

of this paper). On the other hand, (7a) cannot be rejected for the postwar period.[7]

Equation (8) implies that innovations in tax rates should be uncorrelated with all innovations in returns. This is a prediction of the tax-smoothing model that, to my knowledge, has not been tested before. For the implementation of the test, notice that all linear combinations of security returns—differences between pairs of returns in particular—must also be uncorrelated with tax changes and that fixed components of returns do not matter for return innovations. Any real return is the difference of the nominal return and inflation. The innovation in inflation is -1 times the real return innovation of a one-period nominal bond. Thus, provided a one-period nominal bond (a three-month Treasury bill) is in the set of assets, equation (8) implies zero covariance between innovations in tax changes and all nominal returns. Inflation data are not needed. Using linear combinations and dropping known components are also useful for cases in which returns have several parts (e.g., for bonds with different maturities and for foreign-currency investments).

Covariances between innovations of tax rates and returns were computed for a variety of widely available securities. The return series are the change in three-month Treasury bill yields (symbolized DTB), the nominal return on long-term Treasury bond (LRET), nominal stock returns measured by the Standard and Poor's (S&P) 500 index (STOCK), and changes in German and Japanese exchange rates (EG, EJ), money-market yields (SG, SJ), and long-term government bond yields (LG, LJ).[8] Yield changes in nominal bonds are used as proxies for capital gains, which are the uncertain part of bond returns.[9] Exchange rates capture the risky component of the return on a three-month foreign money-market investment. Changes in foreign yields measure the difference between long- and short-term foreign bond-market investments. (Though some of the series represent linear combinations or only the uncertain components of returns, I will refer to them as "returns.") Finally, to test whether indexed debt might improve tax smoothing, the innovation in inflation (GNP-deflator; symbolized P) is considered as a potential return series.

Innovations in returns and tax rates were computed from vector autoregressions (VAR) of tax rates and returns, generally with four lags. For each return series, Table 1 displays the correlation, ρ_k, between innovations in taxes and the return, the covariance denoted by $c_k = E_0(\hat{\tau}_{t+1}\hat{r}_{t+1,k})$ and its asymptotic standard error denoted by std-c_k. In addition, Table 1 indicates (by + symbols) whether an ordinary least-squares regression of tax rates on the current return and lagged values of both series has a significant coefficient on the current return. This test is motivated by the fact that the zero-covariance restriction (8) implies a value of zero for this regression coefficient.

As Table 1 shows, tax smoothing is clearly rejected with both domestic bond series, with stock returns, with the inflation se-

[7]Sahasakul (1986) examines contemporaneous relations between tax rates and other government-sector variables instead of the predictability of tax rate changes, and he uses marginal income-tax rates instead of the revenue:GNP ratio as basic data series. He finds a positive relation between temporary military spending and tax rates, which indicates a violation of equation (7a). Using the data provided in his paper, I find that tax rates are indeed not a random walk over his sample period (1937–82); they are positively autocorrelated. However, for the postwar period (1954–82), (a) the positive relation between temporary military spending and tax rates vanishes, and (b) one cannot reject that tax rates follow a random walk. A Chow test indicates a significant break in the autocorrelation coefficient.

[8]Data sources are national income accounts for macroeconomic series, the International Monetary Fund for international data, the Center for Research in Security Prices (CRSP) for Treasury-bill yields, and Ibbotsen Associates for stock and bond returns. Domestic returns are based on the last day of a quarter, and international returns are based on the last month of a quarter. Details are available from the author.

[9]Because of changing duration, this is only an approximation for finite intervals. For the Treasury-bill market, exact three-month holding returns on six-month bills were available from 1959 on. Estimated results for return innovations computed from yield changes and from holding returns were very similar. Thus, the longer series on yield changes was used throughout the study.

TABLE 1—CORRELATIONS WITH TAX RATES

Return series	ρ_k	c_k	std-c_k
A. 1954:2–1987:4:			
DTB	−0.201	−0.243	0.106**++
LRET	−0.281	−0.627	0.206***+++
STOCK	−0.196	−0.646	0.289**++
P	−0.294	−0.051	0.016***+++
B. 1973:1–1987:4:			
EG	0.091	0.276	0.393
EJ	0.019	0.056	0.377
SG	−0.223	−0.029	0.017*
LG	−0.431	−0.031	0.010***+++
SJ	−0.255	−0.027	0.014*+
LJ	−0.141	−0.096	0.088

Legend: ρ_k is the correlation between return series k and the tax rate, c_k is the covariance between return series k and the tax rate, and std-c_k is the asymptotic standard error of c_k.

Notes: Stars indicate rejection of H_0: $c_k = 0$ at the 10-percent (*), 5-percent (**), or 1-percent (***) significance level based on the asymptotic standard error std-c_k. Plus signs (+, ++, or +++) indicate rejection of the same hypothesis at the same, respective, significance levels, based on a regression of tax rates on current returns and four lags of both series.

ries, and with some of the international series. That is, the government would have been able to improve tax smoothing by having different quantities of the corresponding securities in its portfolio of liabilities.[10] Unfortunately, these statistical rejections provide little information on how policy could be improved, and they may be sensitive to institutional rigidities in the tax-setting process.[11] The next section will therefore explore how the optimal structure of government liabilities can be computed explicitly.

III. The Optimal Structure of Government Liabilities

To obtain a solution for the optimal debt structure, the link between innovations in debt and tax rates must be made explicit (i.e., a formula for innovations in tax rates, $\hat{\tau}_{t+1}$, is needed). Let y_t be the growth in output (= endowments; empirically, GNP), let $\bar{y}$ be the mean, and assume $\bar{y} < r$. Denote the new information about period-$(t+j+1)$ output growth by $\hat{y}_{t+1+j} = E_{t+1}y_{t+1+j} - E_t y_{t+1+j}$, denote the new information about period-$(t+j+1)$ government spending relative to output by $\hat{g}_{t+1+j} = (E_{t+1}G_{t+1+j} - E_t G_{t+1+j})/Y_t$, and let $d_{t,k} = p_{t,k}D_{t,k}/Y_t$ be the ratio of security-k debt to output. Then, an approximate solution for the period-$(t+1)$ innovation in tax rates is

$$(9)\quad \hat{\tau}_{t+1} = \tau_{t+1} - E_t\tau_{t+1}$$
$$= (1-\psi)\cdot\exp(-\bar{y})$$
$$\times\left[\sum_k \hat{r}_{t+1,k}d_{t,k} + \sum_{j\geq 0}\rho^j\hat{g}_{t+1+j}\right]$$
$$-\tau_t\sum_{j\geq 0}\psi^j\hat{y}_{t+1+j}$$

[10]To verify that the results are not due to misspecification of the regressions or the assumed availability of a risk-free asset, a test based on equation (7b) was implemented by testing whether the product series $\Delta\tau_{t+1}(r_{t+1,k} - r_{t+1,l})$ has a zero mean for any pair of security returns (k,l). [The product $\Delta\tau_{t+1}(1+r_{t+1,k})$ was not used directly, because its mean is dominated by the $\Delta\tau_{t+1}$ component; a test would differ little from testing (7a).] Series LRET and STOCK, for which complete return data are available, were taken as securities k, the three-month Treasury-bill as l. The t statistics of −3.71 and −2.00 confirm the rejections reported in Table 1 at the same levels of significance (1 percent and 5 percent, respectively). Since the simple tests already yield strong rejections, more elaborate tests (e.g., following Lars Hansen and Kenneth Singleton [1983]) seem unnecessary.

[11]It may be tempting to use the correlations in Table 1 to assess the economic significance of the rejections. The fact that most correlations are below 0.3 in absolute value (except for LG) suggests that the optimal supply of any one security would have reduced the variance of tax rates by less than 10 percent (18.6 percent for LG; 11.7 percent for all four securities in Part A of Table 1 jointly, estimated using a four-lag VAR with tax rates and all four return series). However, as noted earlier, one should be cautious in interpreting the results based on high-frequency tax-rate data. If there are short-term institutional rigidities in tax policy, marginal changes in the liability portfolio (to optimize debt management) might not translate into changed tax policy on a quarterly basis. The tests show that the variance of tax rates has not been minimized, but they do not reveal what is responsible for the excessive variance.

where $0 < \psi = \rho \cdot \exp(-\bar{y}) < 1$ is a discount factor.

The derivation is conceptually straightforward but lengthy.[12] The idea is that the present value of tax revenues must cover initial debt plus the present value of spending. Any innovation in current or future government spending and any unexpected change in the value of debt therefore forces the government to adjust tax revenues eventually. Because of tax smoothing, a fraction $(1-\psi)$ of the adjustment takes place immediately. In addition, since the present value of tax revenues depends on the path of output, tax rates have to be changed whenever new information about current or future output is received. As a result, tax rates are increased if the value of debt increases unexpectedly, if estimates of future government spending are revised upwards, or if output is lower than expected.

It may be worth noting that this argument applies even if tax rates cannot be adjusted every period. Then, optimal debt policy would have to stabilize the Lagrange multiplier of the budget constraint, which would replace the marginal excess burden $h'(\tau)$ in the first-order condition (8). This multiplier would be perfectly correlated with the right-hand side of (9), leaving the results for optimal debt structure unchanged.

Using the tax-rate formula (9) in the first-order condition (8), one obtains a system of K equations for the K "risky" securities, $d_{t,k}$, issued by the government:

$$\sum_{l} \mathrm{Cov}_t(\hat{r}_{t+1,l}, \hat{r}_{t+1,k}) d_{t,l} + \mathrm{Cov}_t\left(\hat{r}_{t+1,k}, \sum_{j \geq 0} \rho^j \hat{g}_{t+1+j}\right) - w_t \mathrm{Cov}_t\left(\hat{r}_{t+1,k}, \sum_{j \geq 0} \psi^j \hat{y}_{t+1+j}\right) = 0$$

for all l, where $w_t = [\exp(\bar{y})/(1-\psi)]\tau_t$ is a weighting factor. To simplify, let $\mathbf{d}_t$ be the vector of risky government debt securities $d_{t,l}$, let $\mathbf{\Sigma}_r$ be the variance-covariance matrix of returns, assumed to be nonsingular, and let $\mathbf{\Sigma}_{g,r}$ and $\mathbf{\Sigma}_{y,r}$ be the vectors of covariances between returns and the present-value expressions $\sum_{j \geq 0} \psi^j \hat{y}_{t+1+j}$ and $\sum_{j \geq 0} \rho^j \hat{g}_{t+1+j}$, respectively. Then, the above equation can be restated as $\mathbf{\Sigma}_r \mathbf{d}_t + \mathbf{\Sigma}_{g,r} - w_t \mathbf{\Sigma}_{y,r} = 0$, and the optimal debt structure is

$$\mathbf{d}_t = w_t \mathbf{\Sigma}_r^{-1} \cdot \mathbf{\Sigma}_{y,r} - \mathbf{\Sigma}_r^{-1} \cdot \mathbf{\Sigma}_{g,r}. \tag{10}$$

Thus, the general formula for optimal debt structure involves covariances of returns with innovations in output and government spending.

Output (or equivalently, aggregate income) matters in this model, because it forms the tax base and because high tax *rates* cause distortions. Given desired *levels* of revenues, tax rates must increase if output falls. Since tax rates are smoothed over time, any news about future output is also relevant. Consequently, permanent changes in output have much larger effects than temporary changes. Unexpectedly high government spending has an additional effect on tax rates. The optimal debt structure is chosen to hedge against these sources of uncertainty and thereby minimizes fluctuations in tax rates.

IV. Estimates of the Optimal Debt Structure

In this section, formula (10) will be used to explore the optimal structure of government liabilities.

A. *Methodology*

Equation (10) identifies uncertain output and uncertain government spending as sources of risk. Since the two covariance vectors enter as weighted sums in this equation, the optimal debt structure can be interpreted as the sum of two components. A priori, it seems likely that output variation is a quantitatively significant source of risk; the cyclical volatility of budget deficits is

[12] Details are available from the author. Quadratic excess burden $h(\tau)$ is assumed, and the growth rate of output, y_t should be stationary but not necessarily independently and identically distributed.

well documented. Interesting correlations of returns with output (GNP) were indeed found, while correlations with spending turned out to be small and largely insignificant. (Details are in an earlier version of this paper, which is available on request.) In reporting empirical results, I will therefore focus on the output component. This is done formally by imposing the auxiliary assumption that correlations between government spending and debt are approximately zero, $\Sigma_{g,r} = 0$. Then, the optimal structure of debt is proportional to the vector $\mathbf{s} \equiv \mathbf{\Sigma}_r^{-1} \cdot \mathbf{\Sigma}_{y,r}$, which depends only on the variance-covariance matrix of innovations in output and security returns, $\mathbf{d}_t = w_t \mathbf{s}$.

It is useful to keep the factor of proportionality, w_t, separate because w_t depends critically on the discount rate applied to future output. For a real discount rate of $1-\psi = 1$ percent per quarter, $\tau_t \cong 0.2$, and $\exp(\bar{y}) \cong 1$, the proportionality factor is $w_t = [\exp(\bar{y})/(1-\psi)]\tau_t \cong 20$; but if $1-\psi = 0.5$ percent were assumed, the factor would double. Recalling that $\mathbf{\Sigma}_{y,r}$ is the covariance between the *present value* of output and returns, the discount factor also enters into the vector $\mathbf{s}$, but it turns out that different discount rates have a negligible effect on the estimates. (To save space, results for $\mathbf{s}$ will be shown for $\psi = 0.99$ only.)

Thus, I will concentrate on computing the vector $\mathbf{s}$, which indicates whether securities enter with positive or negative sign into the optimal portfolio and in which relative quantities. For the interpretation, a range of "reasonable" weights, say between 10 and 40, may be applied to compute $\mathbf{d}_t$. The main econometric problem in estimating $\mathbf{\Sigma}_{y,r}$ or $\mathbf{s}$ is to identify the innovations, in particular the change in expectations of "far out" realizations of output growth, which enter $\mathbf{\Sigma}_{y,r}$ through the expected-present-value expression $\Sigma_{j \geq 0} \psi^j \hat{y}_{t+1+j}$. Vector-autoregression (VAR) techniques were used because they seem ideally suited for the tasks of extracting the covariance structure and computing projections of a multivariate process.

For all securities, I started with a minimal bivariate VAR including only GNP growth (for y_t) and a single return series, using quarterly U.S. data from 1954:2 to 1987:4, including a constant and four lags. Several alternative versions were estimated to see whether the results are robust to changes in specification.[13] Finally, I computed optimal debt structures for several return series simultaneously, with and without additional information variables. These alternative specifications will only be displayed when they affect the conclusions from the basic bivariate process.

Consistent point estimates and asymptotic standard errors of the elements of $\mathbf{\Sigma}_{y,r}$ and $\mathbf{s}$ (denoted by c_k and s_k in the tables) can be obtained as functions of the VAR coefficients and the residual variance-covariance matrix (see Theodore Anderson, 1958; Takeshi Amemiya, 1985; Peter Schmidt, 1976). If the VAR includes only y_t and a single return series, the signs of c_1 and h_1 can also be determined with a simpler auxiliary regression of y_t on the current return and lagged values of both series. Details are available from the author upon request.

B. *Domestic Debt Securities*

Currently all U.S. government liabilities are nonindexed dollar-denominated debt securities with various maturities. To focus on indexation first, consider a debt portfolio with only two securities, an indexed bond ($k = 0$) and a one-period nominal bond ($k = 1$). Since the real return on a one-period nominal bond is the known promised yield, the innovation in real return is -1 times the rate of inflation. Thus, the covariance between the present value of GNP and inflation determines whether nominal debt is desirable as a hedge. Estimates based on VAR's with GNP growth and inflation are displayed in Table 2 (where all estimates

[13]The alternative estimates used eight instead of four lags, sample periods 1954:2–1972:4 and 1973:1–1978:4 (intended to capture potential breaks in the processes), or one or more additional variables in the information set (other returns, military spending, money supply M1, the monetary base, and import prices). All macroeconomic variables are log-differenced.

TABLE 2—RETURNS ON NOMINAL BONDS

VAR	ρ_k	c_k	std-c_k	s_k	std-s_k
1	0.853	0.511	0.254**++	3.28	1.59**
2	0.534	0.240	0.102**	1.95	0.80**
3	0.540	0.237	0.096**	2.00	0.77***
4	0.461	0.166	0.075**	1.63	0.70**
5	0.388	0.133	0.069*	1.34	0.68**

Legend: ρ_k is the correlation between a return series and the present value of output; c_k is the covariance between a return series and the present value of output; std-c_k is the asymptotic standard error of c_k; s_k is the indicator of optimal supply of a security, as defined in Section IV-A; and std-s_k is the asymptotic standard error of s_k.

Notes: The columns of c_k and std-c_k have been multiplied by 10^4 to improve readability. Stars indicate rejections of $c_k = 0$ or $s_k = 0$ at the 10-percent (*), 5-percent (**), or 1-percent (***) significance level based on the asymptotic standard errors (Wald tests). Plus signs (+, + +, or + + +) indicate rejection of the same hypotheses at the same, respective, significance levels, based on a regression of GNP growth on current returns and four lags of both series. The VAR specifications are:

1) bivariate with GNP growth and inflation, sample 1954:2–1987:4, four lags;
2) with money supply M1, money base, and DTB as information variables; otherwise as for VAR 1;
3) with M1, money base, military spending, and DTB as information variables; otherwise as for VAR 1;
4) with M1, money base, import prices, and DTB as information variables; otherwise as for VAR 1;
5) with M1, money base, import prices, military spending, and DTB as information variables; otherwise as for VAR 1.

have been multiplied by -1; i.e., a positive value means that nominal bonds should be issued).

In the basic specification, VAR 1, the correlation between the innovations is 0.85. This seems extraordinarily high and may reflect the omission of other variables that predict output and inflation; but even if various other variables are included (see VAR's 2–5), the correlations remain around 0.50. The estimates are not only statistically but also economically significant. Taking $s_1 = 1.34$ (the lowest estimate) and a proportionality factor of $w_t = 20$, the ratio of nominal debt (with one-quarter maturity) to GNP should be about $d_1 = 26.8$, as opposed to the current debt:GNP ratio of around 0.50. One has to keep in mind, though, that time-consistency issues are not modeled here, which may reduce the optimal amount of nominal debt. Still, the results suggest that the optimal solution may require the government to hold indexed bonds and to issue nominal debt in an amount far exceeding its total debt. If such simultaneous large long and short positions are impractical, the current policy of issuing only nominal bonds may be interpreted as a corner solution. Alternatively, it may be that the three-month maturity of nominal bonds implicit in quarterly data is too low.

In analyzing the optimal maturity distribution, I will limit the study to the choice between one-period, two-period, and long-term bonds, represented by three-month Treasury bills, six-month Treasury bills, and the longest-term Treasury bonds. Because of the high correlation between interest rates, more variables would probably not add any new insights.

To obtain implications for observable (meaning nominal) return series and to prevent a repetition of the indexation question, it is convenient to restate the term structure in terms of three-month Treasury bills and forward contracts. Given that nominal debt should be issued (as determined above), the question is only whether some of it should have maturities longer than three months. The real return innovation of the long-term bond relative to the three-month bill is given by its nominal return, LRET. The real re-

TABLE 3—MATURITY CHOICE

VAR	ρ_k	c_k	std-c_k	s_k	std-s_k
A. Changes in Treasury-bill rates:					
1	0.084	0.268	0.590	0.38	0.83
2	0.568	1.553	0.747**+++	2.41	1.12**
B. Returns on long-term bonds:					
1	0.155	0.70	1.507	0.039	0.060
2	0.334	1.796	1.397	0.074	0.057

Legend: See Table 2 for notation.
Notes: VAR's 1 are bivariate processes with GNP growth and a return series (DTB or LRET), using sample data for 1954:2–1987:4, including four lags. VAR's 2 use eight lags instead. The columns of c_k and std-c_k have been multiplied by 10^5 in Part A and by 10^4 in Part B to improve readability.

turn on a six-month, relative to a three-month, Treasury bill is the six-month yield at the beginning of the period minus the three-month yield at the end of the period. Since six-month yields are not available for the entire sample, the return innovation is proxied by the change in three-month yields, denoted by DTB (For 1960–87, results were similar to those with the exact return series and are therefore not reported.)

Innovations in LRET and DTB can be interpreted as return innovations on a three-month forward contract on the security; that is, correlations with the present value of output determine whether the government has a hedging demand for forward contracts. An optimal short position together with the supply of three-month Treasury bills would establish the optimality of issuing longer-term bonds.

Estimates are displayed in Table 3. All correlations between the present value of output with DTB and LRET have positive signs, which indicates that two-period or long-term nominal bonds (or equivalently, forward contacts) should be issued. Unfortunately, only the estimate for DTB based on an eight-lag VAR is significant. All the positive correlations seem to be due to a delayed reaction of output to interest rates, which comes out strongest in processes with long lags. Similar results were obtained when the optimal supply of three-month, six-month, and long-term bonds (or two of the three) was estimated jointly as a vector; only the estimate for nominal debt (vs. indexation) was significant. Overall, the point estimates provide some support for issuing long-term debt, but because of the large standard errors, a variety of maturity distributions could be consistent with optimal policy.

C. *Nontraditional Government Liabilities*

There is no reason why governments should restrict their liabilities to nominal or indexed debt securities. In this section, I will consider two other classes of securities: stocks and foreign-currency debt.

German mark- and Swiss franc-denominated "Carter-bonds" were issued by the U.S. government in 1978. More recently, yen-denominated debt has been proposed. Concentrating on marks and yen, I consider one-period, two-period, and long-term foreign-currency bonds. Their real returns are linear combinations of nominal exchange-rate changes, nominal interest-rate changes, and domestic inflation. As in the domestic context, it is instructive to analyze the components, which may be interpreted as forward contracts.

Results are displayed in Table 4, all for the period of flexible exchange rates 1973:1–1987:4. The return series are the rates of dollar depreciation relative to marks and yen (EG and EJ, respectively) and -1 times the change in short- and long-term German and Japanese interest rates (SG and LG for Germany, SJ and LF for Japan, respectively).

TABLE 4—OPTIMAL PORTFOLIOS WITH FOREIGN-CURRENCY-DENOMINATED SECURITIES

RET	ρ_k	c_k	std-c_k	s_k	std-s_k
A. One-period U.S. dollar-bonds and one- and two-period mark bonds:					
P	0.796	0.387	0.285	3.264	2.239**
EG	0.324	2.373	2.154	0.079	0.064
SG	0.251	0.076	0.075	2.606	1.587
B. One-period U.S. dollar and mark bonds and long-term mark bonds:					
P	0.361	0.154	0.250	0.626	1.927
EG	0.340	2.286	1.844	0.133	0.960**
LG	0.544	0.081	0.051	7.258	3.215**
C. One-period U.S. dollar-bonds and one- and two-period yen bonds:					
P	0.699	0.291	0.184	2.333	1.515
EJ	−0.191	−1.312	2.094	−0.036	0.063
SJ	0.795	0.188	0.075**	4.657	1.629***
D. One-period U.S. dollar and yen bonds and long-term yen bonds:					
P	0.540	0.226	0.177	1.97	0.364
EJ	0.057	0.355	1.896	0.070	0.076
LJ	0.598	0.091	0.048*	7.528	3.526**

Legend: See Table 2 for notation.
Notes: Each panel is estimated with a four-variable VAR with GNP and the three return series listed under RET, using sample data from 1973:1–1987:4, including four lags. The columns of c_k and std-c_k have been multiplied by 10^4 to improve readability.

Since the question of whether there is an incremental benefit in issuing German or Japanese currency bonds arises in the presence of domestic nominal bonds, I concentrate on the multivariate framework. In Part A of Table 4, nominal domestic bonds (with real returns contingent on $-1 \times$ inflation, P) are considered jointly with one- and two-period German bonds. The quarterly exchange-rate movements, EG, indicate the return on three-month investments in Germany relative to Treasury bills. The change in German interest rates, SG, proxies the return on six-month relative to three-month investments. The positive correlations and point estimates suggest that both variables may have hedging roles, but they are insignificant.

Significant positive results were obtained by the analogous regressions with long-term German bonds in Part B of Table 4 and with short- and long-term Japanese bonds in Parts C and D. (Bivariate VAR's with GNP growth and a single return series were similar: estimates for LG, SJ, and LJ are significant.) Interestingly, the optimal exposure to yield changes exceeds the optimal total exposure to exchange-rate risk in all cases.[14] The optimal hedge would have to combine short positions in short-term foreign securities with larger holdings of longer-term bonds. Overall, exposure to selected foreign interest rates appears to be desirable. Though a more comprehensive analysis of foreign-currency debt is beyond the scope of this article, there seems to be some potential for improvements in United States debt policy in this direction.

Finally, stock prices are commonly considered to be highly cyclical variables, which makes them natural candidates for hedging output risk. Given that nominal debt is pre-

[14]Exposure to exchange-rate risk may be desirable for reasons not considered here (e.g., for providing incentives or credibility in the context of exchange-rate stabilization).

TABLE 5—OPTIMAL PORTFOLIOS OF NOMINAL BONDS AND STOCKS

RET	ρ_k	c_k	std-c_k	s_k	std-s_k
A. Nominal bonds and stocks:					
P	0.847	0.491	0.253**	2.932	1.652**
Stocks	0.451	5.073	2.023	0.050	0.034
B. Nominal bonds and stocks (five-variable VAR):					
P	0.660	0.334	0.150**	2.031	1.035**
Stocks	0.530	5.198	1.876***	0.074	0.033**

Legend: See Table 2 for notation.
Notes: Part A is estimated with a three-variable VAR with GNP, inflation, and stock returns, using sample data from 1954:2–1987:4, including four lags. The VAR for Part B includes M1 and the monetary base as additional variables. The columns of c_k and std-c_k have been multiplied by 10^4 to improve readability.

sent, optimal nominal debt and the optimal stock market position were estimated jointly in Table 5. Part A is based on a VAR with output growth, inflation, and stock returns (measured by the S&P 500 index). Part B is based on a VAR that includes the M1 money supply and the monetary base as additional variables. In both cases, the optimal debt structure includes a short position in stocks, which is significant in the larger process that is based on the better estimate of inflation. (Similar estimates were obtained in a bivariate VAR with GNP growth and stock returns only, which were significant at the 1-percent level.)

To my knowledge, a proposal suggesting government participation in the stock market has not been made before. Based on the statistical evidence linking stocks to the present value of output, a short position would provide a hedge against cyclical shortfalls in government revenue. However, when exploring such nonstandard financing strategies, one may ask why the government should not go one step further and sell synthetic securities that are directly contingent on GNP. A theory of market incompleteness (beyond the scope of this paper; see Gale, 1990) would be needed to decide whether such securities could be issued. However, the empirical evidence suggests that innovative financing strategies, with existing or synthetic securities, are worth exploring.

V. Conclusions

The optimal structure of government debt has been analyzed in a stochastic environment. In a setting with distortionary taxes, the government should smooth tax rates over states of nature as well as over time. This requires state-contingent government liabilities that provide a hedge against shocks to the budget.

For postwar U.S. data, tax smoothing as a positive theory of policy cannot be rejected on the basis of the time path of taxes. However, a number of security returns are correlated with tax rates, leading to a rejection on that basis. Estimates of optimal debt portfolios provide strong support for using nominal, nonindexed, government debt, but provide only weak evidence on the maturity distribution. Moreover, it seems that the government could improve tax smoothing by having some nontraditional liabilities, like foreign-currency debt or a short position in the stock market.

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